Fluid Flow and Heat Transfer in a Double-Pipe Heat Exchanger

Aim: To analyze conjugate heat transfer between hot and cold laminar streams separated by a conducting wall.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Grid points, along stream(nx)	160
Grid points across stream, cold	20
Grid points across stream, wall	6
Grid points across stream, hot	20
Reynold's number, cold	50, 100, 200
Reynold's number, hot	100
Heat transfer coefficient ratio(solid/fluid)	1,100
Time step	1.25×10^{-3}
Total Iterations	3000

Governing Equation:

$$\rho c_p \, \mathbf{u} \cdot \nabla T = \nabla \cdot (k_f \nabla T).$$

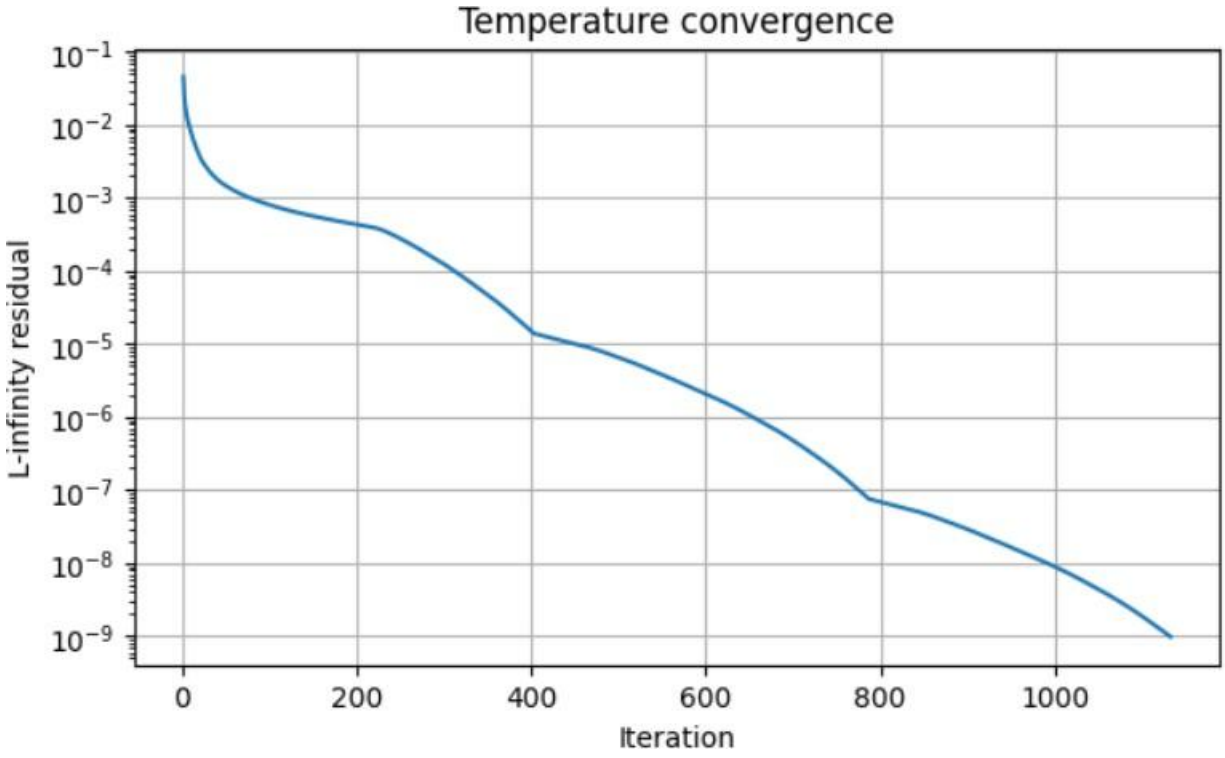
$$\nabla \cdot (k_s \nabla T) = 0.$$

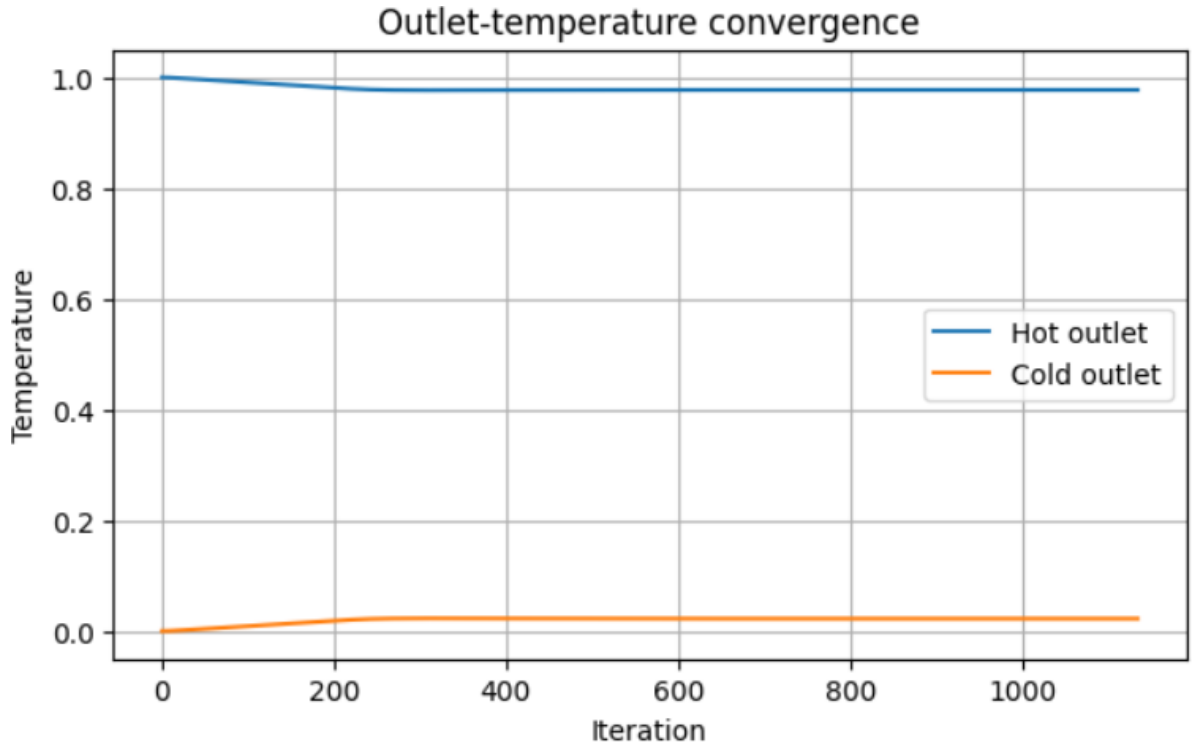
Boundary Conditions: Specified hot and cold inlet temperatures; zero-gradient outlet temperatures; adiabatic external boundaries; continuity of temperature and heat flux at fluid-solid interfaces.

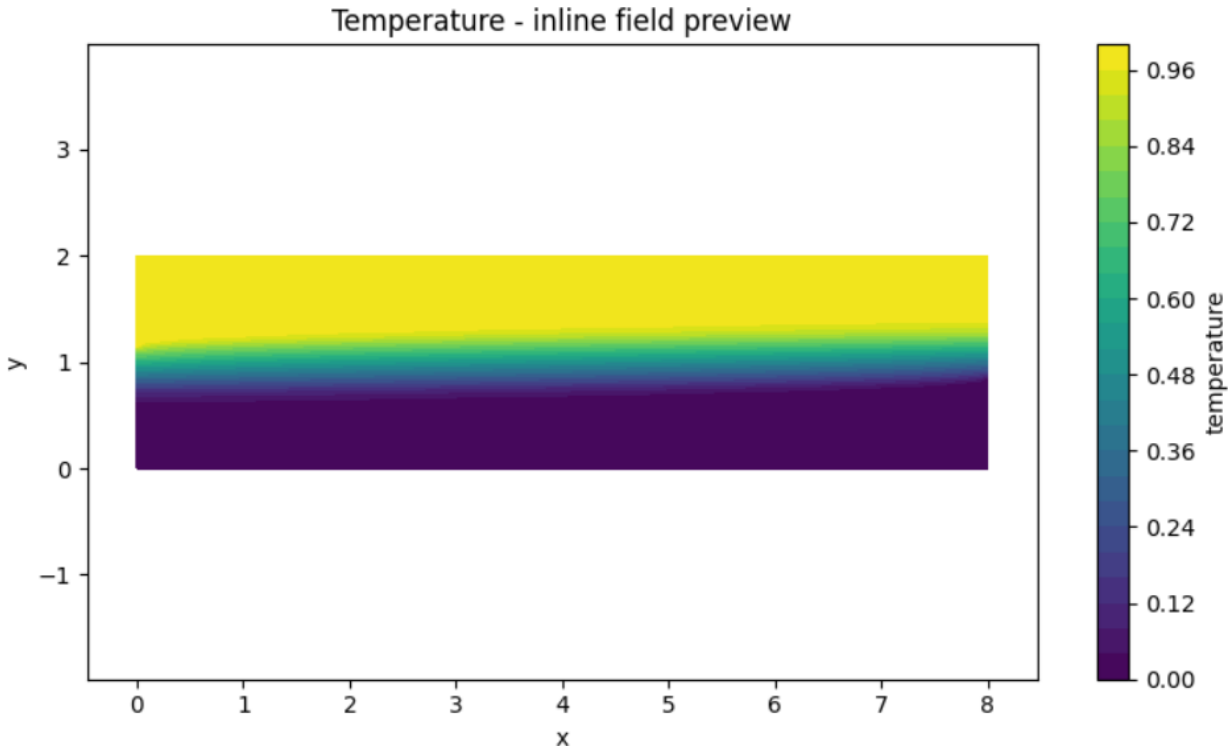
Numerical Method: Steady finite-volume convection-diffusion in two fluid zones coupled to conduction in the intervening wall. Prescribed parabolic velocity profiles isolate the conjugate thermal problem.

Observations: Outlet temperatures, heat-transfer rate, effectiveness, residual, relative energy imbalance, and temperature contours at various Raynold's number for cold fluid. With wall to fluid conductivity ratios 1 and 100.

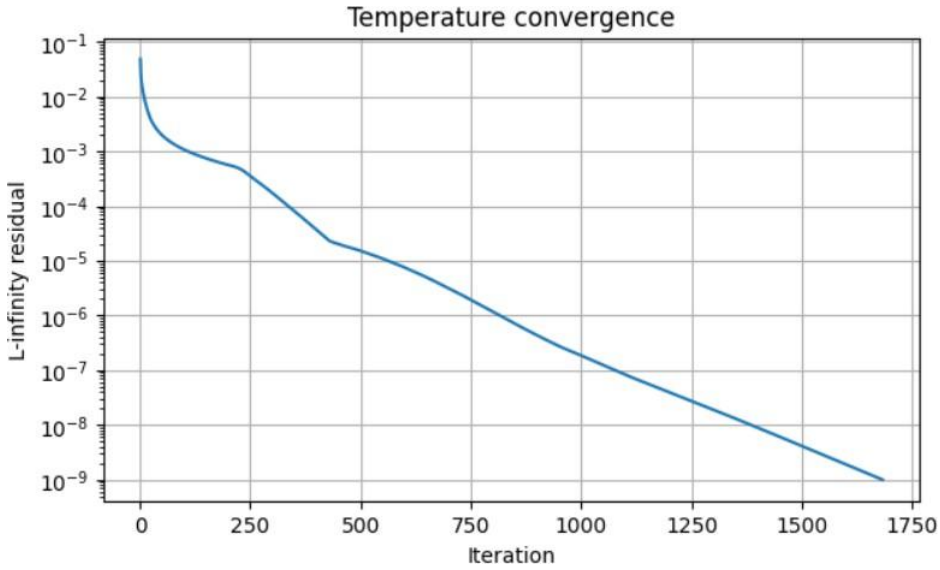
For Counter flow system Raynold's number 50 and wall to fluid ratio 1

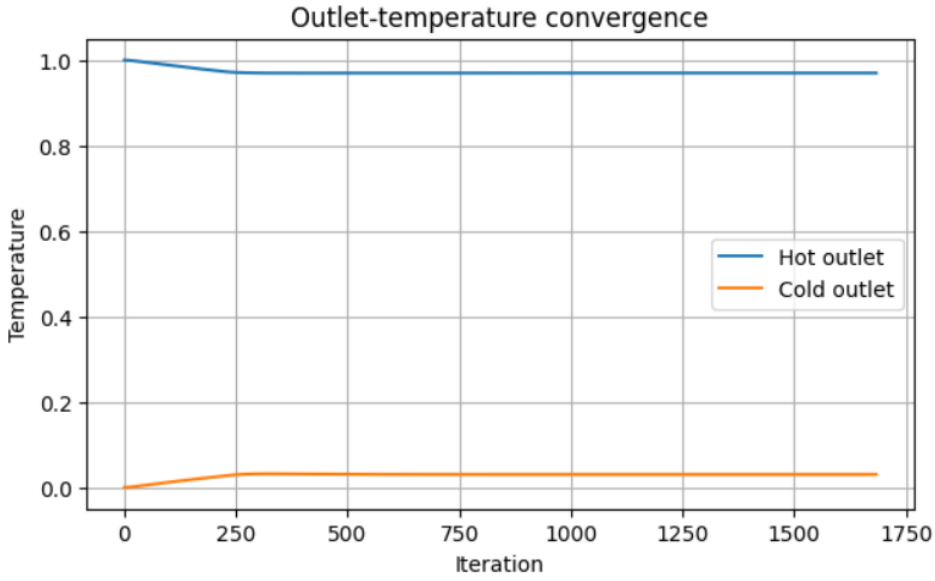


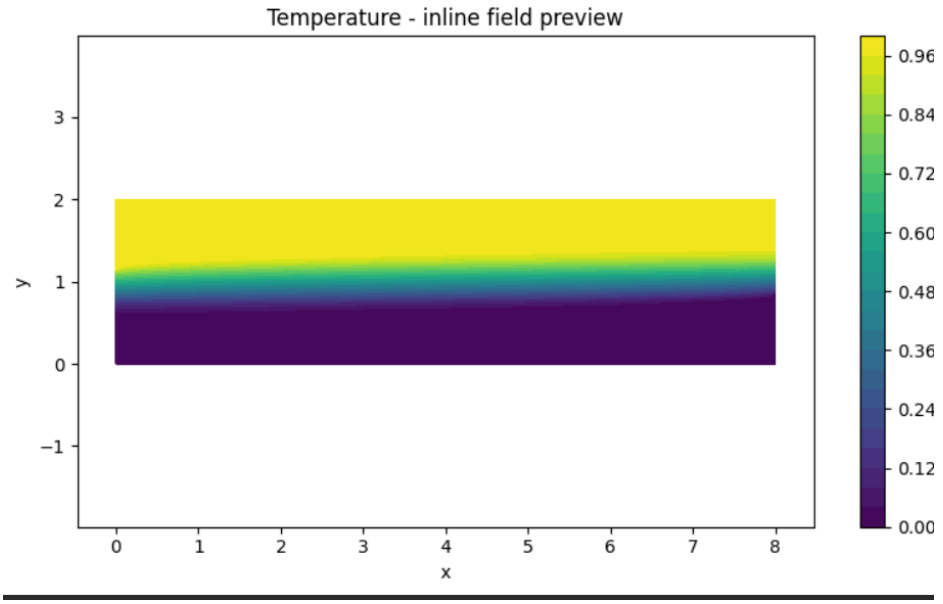




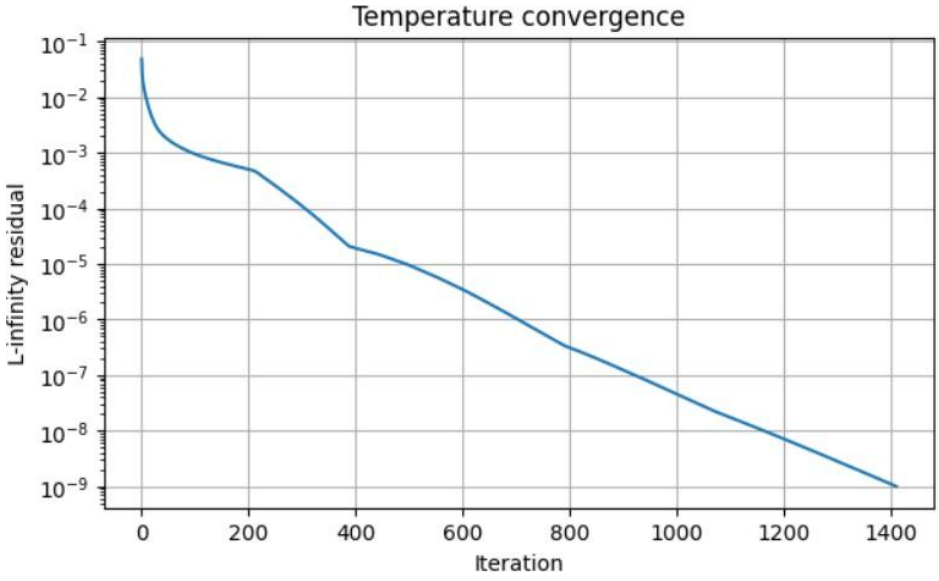
For Counter flow system Raynold's number 100 and wall to fluid ratio 1

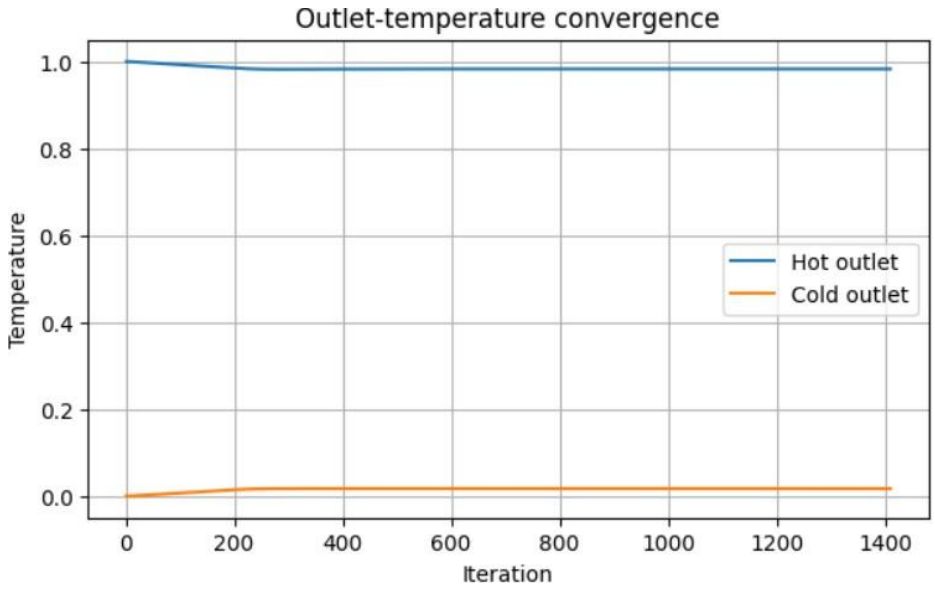


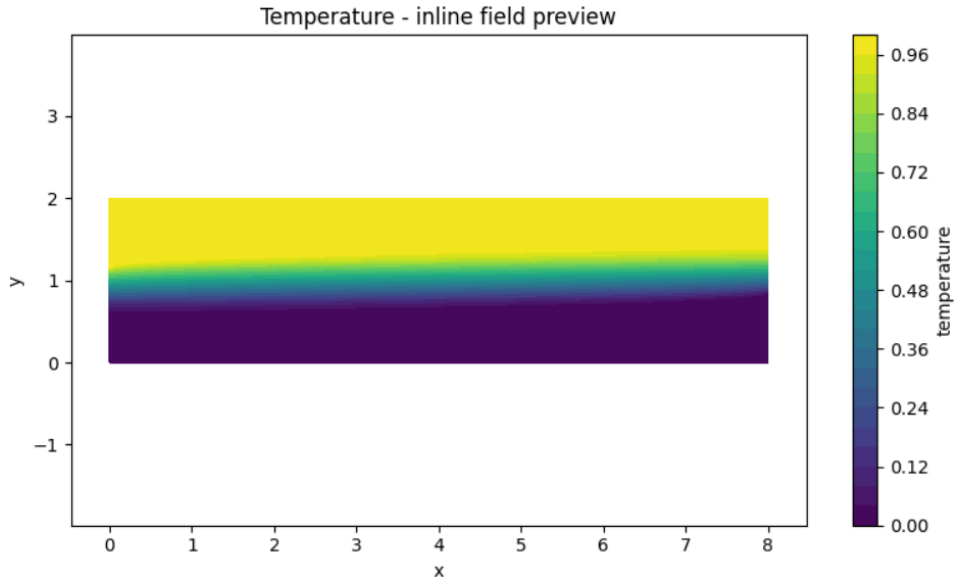

Outlet-temperature convergence
Temperature
Iteration
Hot outlet
Cold outlet
1.0
0.8
0.6
0.4
0.2
0.0
0
250
500
750
1000
1250
1500
1750


Temperature - inline field preview
x
y
3
2
1
0
−1
0
1
2
3
4
5
6
7
8
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

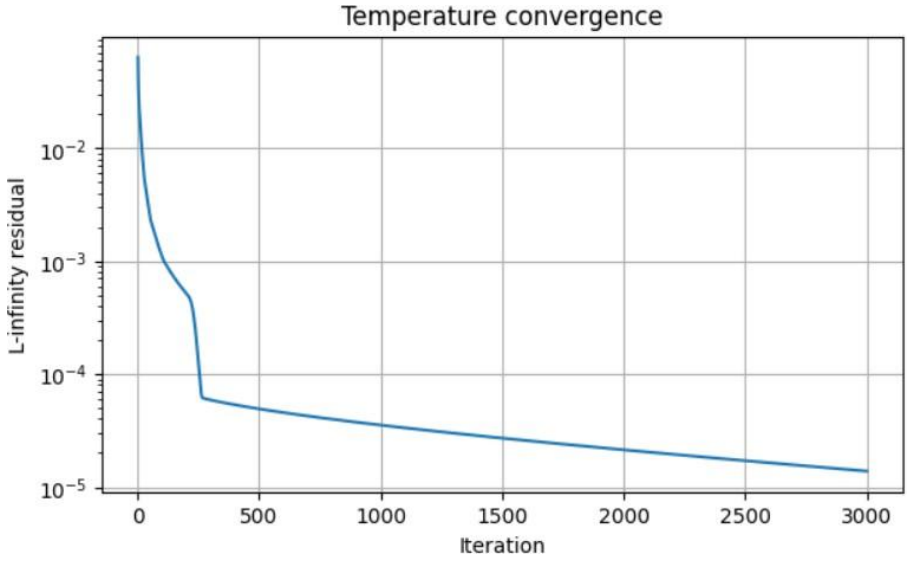
For Counter flow system Raynold's number 200 and wall to fluid ratio 1

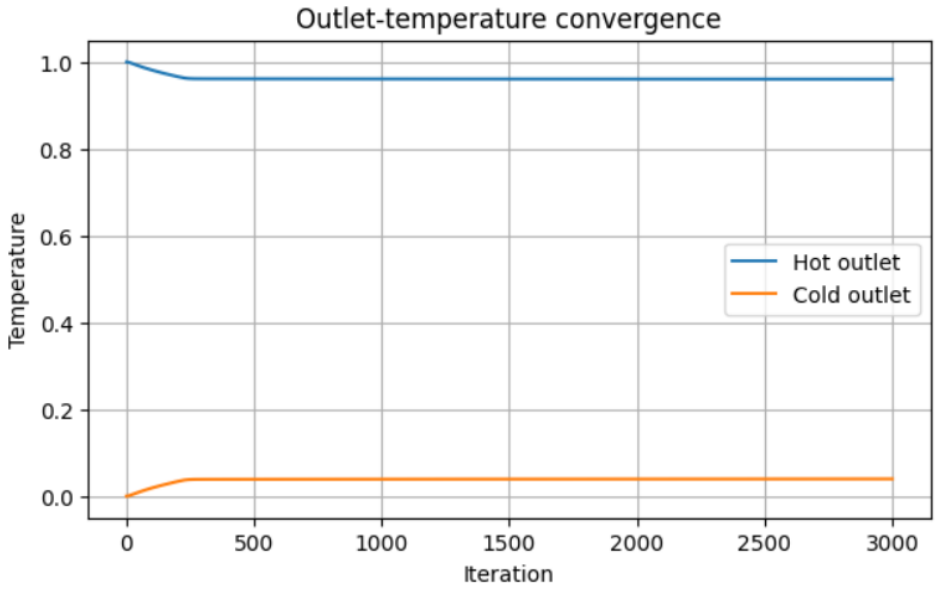


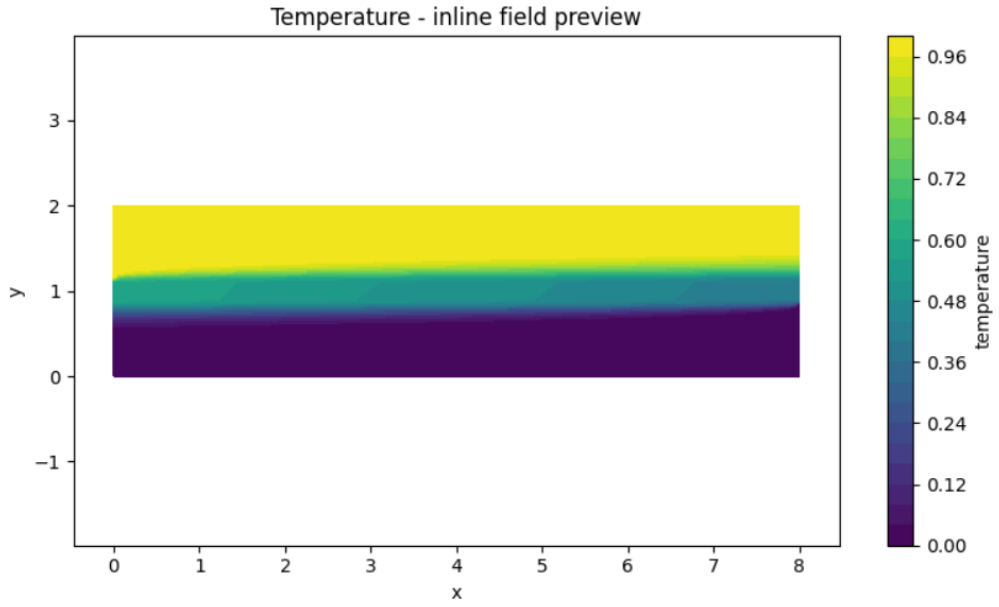




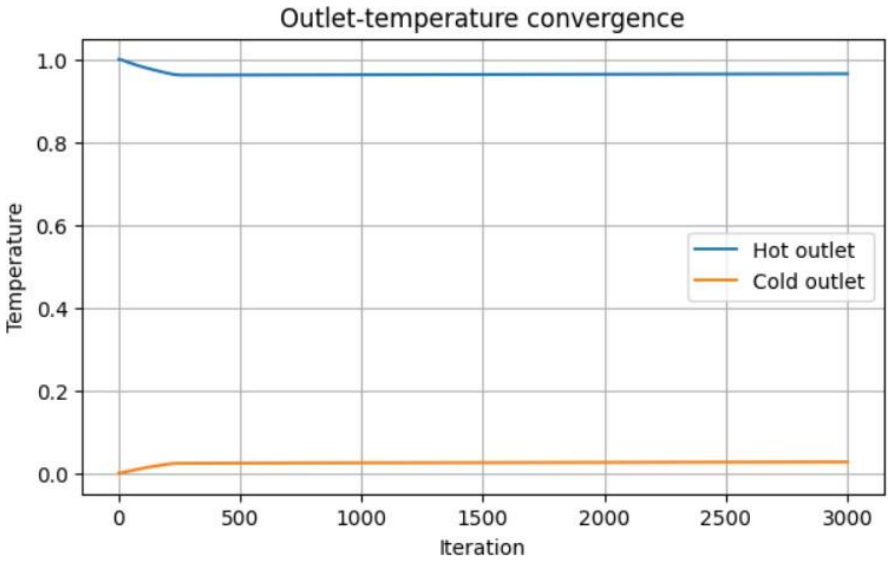
For Counter flow system Raynold's number 100 and wall to fluid ratio 100

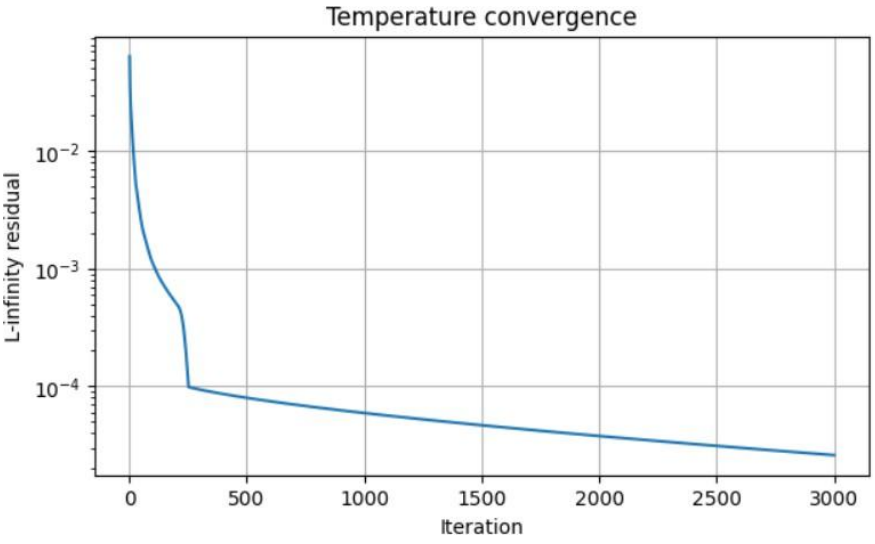


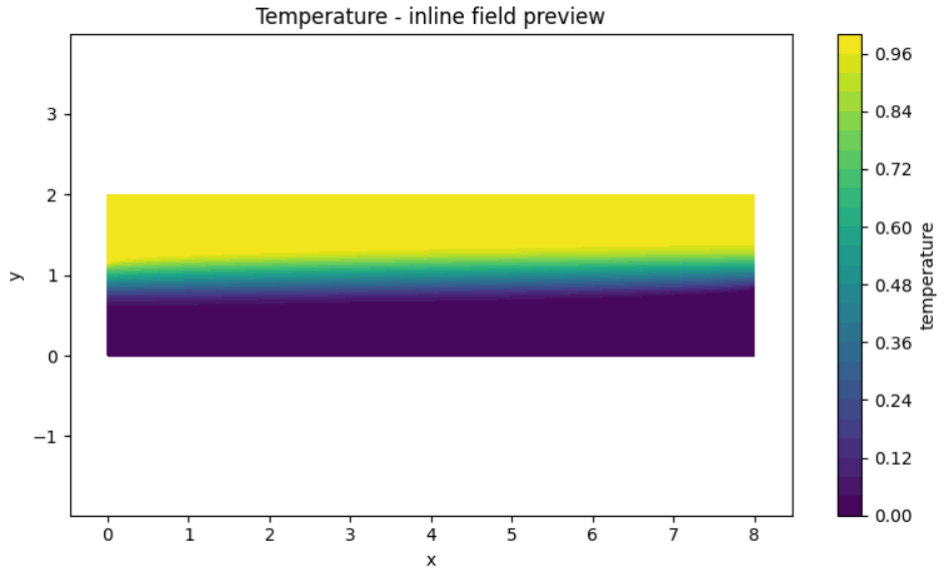

Outlet-temperature convergence
Temperature
Iteration
Hot outlet
Cold outlet
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
2000
2500
3000


Temperature - inline field preview
y
x
temperature
3
2
1
0
−1
0
1
2
3
4
5
6
7
8
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

For Counter flow system Raynold's number 200 and wall to fluid ratio 100







Results:

Reynold's number for cold flow	Ratio of heat transfer coefficient	Hot inlet temperature	Hot Outlet temperature	Cold Outlet temp	Cold Outlet temperature
50	1	1	0.98	0	0.025
100	1	1	0.97	0	0.035
200	1	1	0.98	0	0.020
100	100	1	0.96	0	0.04
200	100	1	0.965	0	0.03

Conclusion:

The hot-stream inlet temperature was maintained at $T_{h,in}$ = 1.0, while the cold-stream inlet temperature was maintained at $Tc_{,in}$ = 0.0. For a wall-to-fluid ratio of 1, increasing the cold-stream Reynolds number from 50 to 100 increases the cold outlet temperature from 0.025 to 0.035, while the hot outlet temperature decreases from 0.98 to 0.97. At $Re_{(c)}$ = 200, the outlet temperatures are approximately 0.98 and 0.020 for the hot and cold streams, respectively. Increasing the wall-to-fluid ratio from 1 to 100 results in enhanced thermal interaction, with the hot outlet temperature decreasing to 0.96 at $Re_{(c)}$ = 100 and 0.965 at $Re_{(c)}$ = 200, while the corresponding cold outlet temperatures increase to 0.040 and 0.030, respectively.